A VANISHING COHOMOLOGY GROUP OF A PSEUDO-COHERENT COMPLEX AT A PRIME CUTS THE LOCALIZED COMPLEX INTO AN UPPER AND A LOWER PART

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author opens the proof by naming a special case of another lemma and then says 'A direct proof is as follows.', supplying that proof in full inside the counted span; the single \ref pointer to lemma-better-cut-complex-in-two is recorded as a non-load-bearing navigation pointer, and the direct argument that replaces it stays inside the counted proof span.

Allowed standard prerequisites (editorial).

- Pseudo-coherent complexes and their representation by complexes of finite free modules. the hypothesis at statement lines 19869-19870 and used at proof lines 19887-19888. Legitimises the reduction to a bounded above complex of finite free modules on which the matrix argument runs.
- Truncations tau_{>= n} and tau_{<= n} of a complex and tor-amplitude. the conclusion at statement lines 19875-19879. Names the two pieces of the asserted decomposition and the property of the upper one.
- Rank of a matrix over a field and nonvanishing of a minor. used at proof lines 19896-19899. Converts the rank statement modulo p into the two minors a and b whose product f is the localizing element.

Source reference. Original lemma 19744: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19744–19744. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $\mathfrak{p} \subset R$ be a prime ideal. Let $i \in \mathbf{Z}$. Let $K^\bullet$ be a pseudo-coherent complex of R-modules such that $H^i(K^\bullet \otimes^{\mathbf{L}}_R \kappa(\mathfrak{p})) = 0$. Then there exists an $f \in R$, $f \notin \mathfrak{p}$ and a canonical direct sum decomposition*

$$K^\bullet \otimes_R R_f = \tau_{\geq i+1}(K^\bullet \otimes_R R_f) \oplus \tau_{\leq i-1}(K^\bullet \otimes_R R_f)$$

in $D(R_f)$ with $\tau_{\geq i+1}(K^\bullet \otimes_R R_f)$ a perfect complex with tor-amplitude in $[i+1, \infty]$.

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Proof. This is an often used special case of Lemma 19744. A direct proof is as follows. We may assume that $K^\bullet$ is a bounded above complex of finite free R-modules. Let us inspect what is happening in degree i:

$$\ldots \to K^{i-2} \to R^{\oplus l} \to R^{\oplus m} \to R^{\oplus n} \to K^{i+2} \to \ldots$$

Let A be the $m \times l$ matrix corresponding to $K^{i-1} \to K^i$ and let B be the $n \times m$ matrix corresponding to $K^i \to K^{i+1}$. The assumption is that $A \bmod \mathfrak{p}$ has rank r and that $B \bmod \mathfrak{p}$ has rank $m - r$. In other words, there is some $r \times r$ minor a of A which is not in $\mathfrak{p}$ and there is some $(m-r) \times (m-r)$-minor b of B which is not in $\mathfrak{p}$. Set $f = ab$. Then after inverting f we can find direct sum decompositions $K^{i-1} = R^{\oplus l-r} \oplus R^{\oplus r}$, $K^i = R^{\oplus r} \oplus R^{\oplus m-r}$, $K^{i+1} = R^{\oplus m-r} \oplus R^{\oplus n-m+r}$ such that the module map $K^{i-1} \to K^i$ kills of $R^{\oplus l-r}$ and induces an isomorphism of $R^{\oplus r}$ onto the corresponding summand of K^i and such that the module map $K^i \to K^{i+1}$ kills of $R^{\oplus r}$ and induces an isomorphism of $R^{\oplus m-r}$ onto the corresponding summand of K^{i+1}. Thus $K^\bullet$ becomes quasi-isomorphic to

$$\ldots \to K^{i-2} \to R^{\oplus l-r} \to 0 \to R^{\oplus n-m+r} \to K^{i+2} \to \ldots$$

and everything is clear. $\square$